SmartSteth: A Convolutional Neural Network (CNN) Driven Auscultation Instrument Using Lung Sound Analysis for Early Respiratory Disease Detection in Satellite Clinics

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Abstract **- Respiratory illnesses such as pneumonia, bronchiolitis, and asthma remain leading causes of morbidity and mortality in rural Philippine communities, where diagnostic resources are scarce and front line health workers often rely on manual auscultation and subjective clinical judgment. This study introduces SmartSteth, a convolutional neural network (CNN)-powered auscultation device designed to enhance early detection of lower respiratory tract diseases in low-resource settings. The device integrates a MEMS microphone with an ESP32 microcontroller and a web-based AI interface to analyze lung sounds in real time, providing immediate, interpretable diagnostic feedback. A labeled dataset of respiratory sounds, including normal, asthmatic, pneumonic, and bronchiolitic recordings, was collected using digital stethoscopes and processed into Mel-spectrograms for CNN training via Edge Impulse. The model underwent iterative validation and was deployed within a web application that receives audio input, performs preprocessing, and displays classification results. Usability testing across four rural clinics showed 100% operability, with health workers successfully navigating the interface and interpreting outputs. The model achieved high classification accuracy, and its offline functionality proved essential in areas with poor connectivity. Users preferred its compact form, waterproof casing, and icon-based interface, while its modular design supported deployment in barangay health stations, mobile clinics, and home visits. SmartSteth demonstrated strong potential to improve clinical decision-making, reduce diagnostic delays, and enhance patient-provider communication. By offering a scalable, low-cost solution for respiratory screening and surveillance, the device supports equitable healthcare access and contributes to key United Nations Sustainable Development Goals, including health, innovation, and reduced inequalities.**

Keywords-*Respiratory diseases, Low-resource settings, AI-powered diagnostics, Web application, Frontline health workers*

I. INTRODUCTION

In the rural barangays of Puerto Princesa City, access to timely and accurate healthcare remains a persistent challenge. Geographic isolation, limited medical personnel, and outdated diagnostic tools hinder early detection and treatment especially for respiratory illnesses that disproportionately affect children, the elderly, and other vulnerable groups. Health workers, often operating with minimal resources, rely on manual auscultation and subjective clinical judgment, which can lead to

misdiagnosis, underreporting, and preventable complications.

Among the most common and concerning respiratory conditions in these communities are pneumonia, bronchiolitis, and asthma. Pneumonia, a serious lung infection, remains a leading cause of mortality in the Philippines. Between January and October 2023, the Department of Health recorded 158,762 pneumonia cases, a 45.68% increase compared to the same period in 2022 (Philstar.com, 2023). In 2024, pneumonia became the fourth leading cause of death nationwide, surpassing diabetes mellitus, with 46,718 reported fatalities (BusinessMirror, 2024). Bronchiolitis, primarily caused by respiratory syncytial virus (RSV), is a leading cause of hospitalization among infants. Although rural-specific data is limited, the disease burden is significant nationwide, and early detection remains critical in areas with limited healthcare access. Asthma, a chronic inflammatory disease of the airways, is frequently underdiagnosed and poorly managed in rural settings. A retrospective cohort study in a rural Philippine site revealed that while 93.3% of children with asthma were prescribed short-acting beta-agonists, only 5.8% received inhaled corticosteroids highlighting a gap in optimal treatment access (PubMed, 2024). Environmental exposures and healthcare access also influence asthma symptoms, with rural residents often experiencing less severe symptoms than their urban counterparts (MIMS, 2023).

These challenges inspired the development of SmartSteth, an AI-powered auscultation device designed to assist frontline health workers in the early identification of lower respiratory tract diseases. By integrating convolutional neural network (CNN) technology with traditional stethoscope functionality, SmartSteth enhances the diagnostic capabilities of health workers without requiring extensive training or technical expertise. The device analyzes lung sounds in real time, providing immediate, data-driven insights that support early identification of respiratory conditions. Its compact form factor and offline functionality make it especially suitable for barangay health stations, mobile clinics, and home visits settings where electricity, internet connectivity, and laboratory support are often unreliable or unavailable.

SmartSteth is designed with the end user in mind. Health workers in remote areas frequently juggle multiple roles, from triage to treatment to data reporting, often under time pressure and with limited supervision. The device's intuitive interface, automated analysis, and visual feedback reduce cognitive load and improve confidence in clinical decision-making. It also facilitates better communication with patients and caregivers by translating complex auscultation findings into understandable results. These features not only improve diagnostic accuracy but also strengthen trust between health workers and the communities they serve.

From a market perspective, SmartSteth addresses a critical gap in the global health

technology landscape: the need for affordable, scalable diagnostic tools tailored to low-resource settings. As governments, NGOs, and health systems increasingly prioritize digital transformation and universal health coverage, there is growing demand for innovations that combine clinical impact with operational simplicity. SmartSteth's modular design and potential for integration with electronic medical records, telemedicine platforms, and community health programs position it as a versatile solution with strong commercial and public health appeal. Its deployment could catalyze broader improvements in respiratory care, data collection, and disease surveillance across underserved regions.

This study aimed to:

1. Evaluate SmartSteth's diagnostic accuracy and usability in rural healthcare settings
2. Understand health worker experiences and community perceptions of AI-assisted screening
3. Identify user-driven design improvements to enhance adoption and effectiveness
4. Explore pathways for market integration and scale-up in remote communities

Without intervention, delayed diagnosis of respiratory illnesses may worsen poverty, strain local health systems, and increase preventable deaths. As highlighted by recent studies and public health reports, strengthening healthcare infrastructure and access in rural areas is essential to addressing the rising burden of respiratory diseases (BusinessMirror, 2023; PubMed, 2024; Philstar.com, 2023). By promoting accessible, tech-enabled screening, this study contributes to key United Nations Sustainable Development Goals (UNSDGs), including Goal 3 (Good Health and Well-being), Goal 9 (Industry, Innovation and Infrastructure), Goal 10 (Reduced Inequalities), and Goal 17 (Partnerships for the Goals). Ultimately, SmartSteth seeks to bridge the gap between innovation and inclusion—empowering health workers and improving respiratory care where it's needed most.

II. MATERIALS AND METHODS

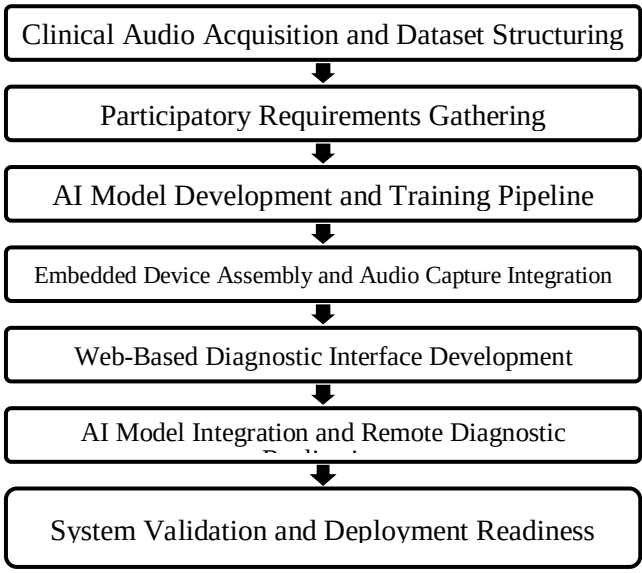


Figure 1. *Schematic Diagram*

Clinical Audio Acquisition and Dataset Structuring

The development of the SmartSteth system began with the acquisition and structuring of a comprehensive lung sound dataset. Audio recordings were sourced from the publicly available Kaggle Respiratory Sound Database, which includes annotated samples of normal breathing and pathological sounds such as wheezes, crackles, and stridors associated with asthma, pneumonia, and bronchiolitis. These recordings were

captured using digital stethoscopes and MAX4466 MEMS microphones, selected for their high sensitivity and low noise characteristics, as supported by Xu and Sankar (2024).

Each recording was labeled according to the patient's clinical diagnosis, forming a supervised dataset suitable for machine learning applications. The dataset was organized to reflect diagnostic categories relevant to lower respiratory tract diseases (LRTDs), enabling the model to learn discriminative acoustic features. This approach aligns with established practices in respiratory sound classification, where annotated datasets are critical for model reliability and clinical relevance (Kim et al., 2022).

Participatory Requirements Gathering

System requirements were defined through participatory engagement with Barangay Health Workers in a Barangay Health Center and Nurses and Staff from Salvacion, Macarascas and San Rafael Satellite Clinics in Puerto Princesa City, Philippines. Using semi-structured interviews and contextual inquiry, the research team identified key diagnostic challenges, including limited access to specialists, time constraints, and the need for intuitive diagnostic tools.

Based on these findings, the hardware configuration was specified to include the ESP32 microcontroller and the MAX4466 MEMS microphone. Software requirements encompassed real-time audio preprocessing, AI model compatibility, and secure data transmission. Functional priorities included rapid classification, offline operability, and a user-friendly interface. However, at this stage of development, SmartSteth operates exclusively as an online system, with classification and result visualization performed via a web-based interface. Offline deployment and embedded integration remain part of future development plans. This user-centered design process reflects principles outlined in the literature on frugal innovation and context-sensitive medical device development (Bhatti et al., 2017; Topol, 2019).

AI Model Development and Training Pipeline

The labeled lung sound dataset underwent preprocessing to enhance signal quality and prepare inputs for model training. Preprocessing steps included denoising, volume normalization, and transformation of audio signals into Mel-spectrograms—a time-frequency representation effective for capturing respiratory sound patterns. These spectrograms were used to train a Convolutional Neural Network (CNN) via the Edge Impulse platform.

Hyperparameters such as learning rate, batch size, and dropout rate were tuned iteratively to optimize classification performance while minimizing overfitting. The model was trained using TensorFlow/Keras with a standard train-validation split, and performance metrics including accuracy, precision, and recall were computed. At present, the trained model is deployed within the online web *application*, where it performs inference on uploaded lung sound data. Prior studies have demonstrated the efficacy of CNNs in respiratory sound classification, particularly when trained on spectrogram representations (Zhang et al., 2023; Pramono et al., 2019).

Assembling of the Device

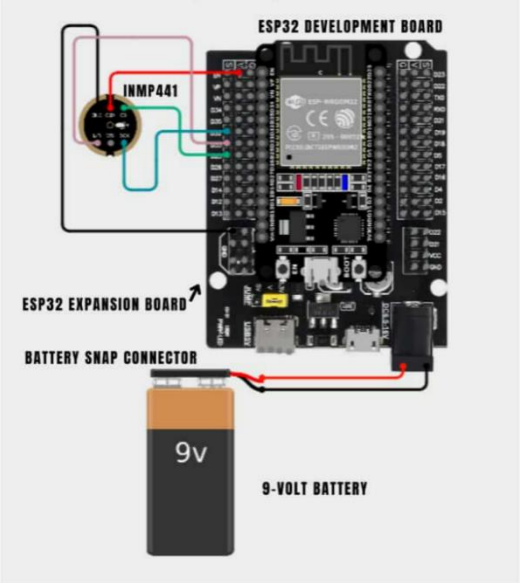


Figure 2. *SmartSteth Cirkit Designer Diagram*

The SmartSteth device was assembled by embedding the MAX4466 MEMS microphone into a standard acoustic stethoscope, enabling digital auscultation. The microphone was connected to the ESP32 microcontroller using jumper wires, with protocol used for digital audio transmission. Pin configurations included VCC to 3.3V, GND to ground, analog OUT to GPIO32.

Power was supplied via USB or a 9V battery with snap connector. The device was housed in a compact enclosure to ensure durability and ease of handling. Functional testing included signal integrity checks, microphone sensitivity calibration, and safety verification. Although the embedded hardware is fully assembled and operational, its integration with the AI model is currently limited to data capture and transmission. Classification and result interpretation are performed online. Full offline deployment of the embedded system is planned for future iterations. This low-cost, modular design reflects principles of frugal innovation in global health technology (Bhatti et al., 2017).

Web-Based Diagnostic Interface Development

A web application was developed to receive, process, and display lung sound data transmitted via Wi-Fi from the SmartSteth device. The backend was implemented using Python and Flask, with support for audio file handling, AI inference, and secure result storage. The frontend was designed for clarity and responsiveness, enabling health workers to upload recordings, view classification results, and playback audio.

The interface was informed by usability heuristics and tailored for low-bandwidth environments. At present, this web-based platform serves as the primary diagnostic interface for SmartSteth, with all classification performed online. Similar web-based diagnostic platforms have demonstrated feasibility in rural health contexts (AshilPrasad, 2023; WHO, 2022).

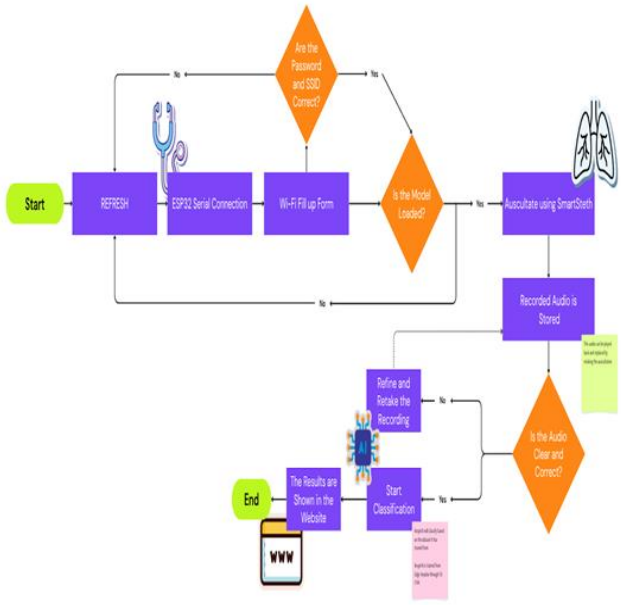


Figure 3. *SmartSteth's Deployment Diagram*

AI Model Integration and Remote Diagnostic Replication

The trained CNN model was exported from Edge Impulse and integrated into the web application using a JavaScript inference library. Uploaded or recorded lung sounds were preprocessed into Mel-spectrograms and passed to the model for classification. The model output included probability scores for each respiratory condition, which were displayed in a user-friendly format.

This integration enables replication of SmartSteth's diagnostic functionality via browser-based tools, expanding accessibility to clinics without embedded hardware. As of this writing, all AI inference is conducted online. Future development will focus on embedding the model within the ESP32 microcontroller to enable offline classification. Literature supports the use of web-integrated AI for scalable diagnostics in low-resource settings (Zhang et al., 2023; WHO, 2022).

System Validation and Deployment Readiness

Final validation involved testing the integrated system, hardware, software, and AI model, for performance and usability. Evaluation criteria included audio quality, Wi-Fi connectivity, classification accuracy, and interface responsiveness. Field simulations were conducted to assess portability, reliability, and user experience in clinical environments.

Feedback from health workers informed minor adjustments to device ergonomics and interface layout. While the system is currently deployed as an online diagnostic tool, its modular design supports future transition to offline use. The validated system is considered ready for pilot deployment in rural clinics, supporting early detection of LRTDs and augmenting frontline diagnostic capacity.

III. RESULTS AND DISCUSSION

Table 1. *Perceived challenges and needs in diagnosing respiratory diseases (LRTDs) among Barangay Health Workers and Satellite Clinic staff*

	Essential Themes	
Category	**Barangay Health Workers (BHWs)**	**Satellite Clinic Staff**

Availability of Diagnostic Tools	*Reliance on basic tools (mainly stethoscope); lack of diagnostic devices limits accuracy.*	*Limited diagnostic equipment; variation in tool quality and availability across facilities.*
Training and Knowledge of Health Workers	*Need for continuous and practical training in lung assessment and symptom recognition.*	*Inconsistent training and skill gaps; need for regular refresher and standardized capacity-building.*
Environmental and Infrastructural Limitations	*Noisy environment and weak signal interfere with assessment; limited physical space.*	*Overcrowded clinics, poor ventilation, and infrastructure strain hinder efficient diagnosis.*
Time Constraints During Consultations	*Heavy patient load; multitasking required; short consultation time per patient.*	*High volume of patients forces quick consultations, affecting diagnostic accuracy.*
Referral Systems and Access to Higher-Level Care	*Referrals limited by patients' financial constraints and travel distance.*	*Hospital congestion and shortage of specialists delay referrals and feedback.*
Community Awareness and Health-Seeking Behavior	*Low health literacy; misconceptions about respiratory symptoms; need for constant community education.*	*Poor early consultation habits; reliance on self-medication before seeking formal care.*

Table 1 revealed that barangay health workers and satellite clinic staff face interconnected challenges in diagnosing respiratory diseases, particularly in barangay-based and rural settings. Barangay workers rely mainly on basic stethoscopes with no access to confirmatory tools, reflecting findings by Ginsburg et al. (2021) that limited diagnostic capacity restricts accurate respiratory assessments in primary care. Many expressed uncertainties in lung sound interpretation, consistent with Rowe et al. (2020), who reported inadequate respiratory training among frontline providers. Environmental noise, limited space, and unstable connectivity further impede auscultation accuracy (Emmanouilidou et al., 2015). In addition, weak referral systems and financial barriers delay escalation to higher care levels (Harrison & Karim, 2019). Low health literacy and misconceptions within communities also reduce early consultation rates (Adebayo et al., 2020). These challenges, especially

prevalent in barangay-based facilities, emphasize the need for accessible diagnostic support and continuous training to strengthen community-level respiratory care.

Table 2. *Frequency of Challenges and Needs in Diagnosing Lower Respiratory Tract Diseases*

Category	User Needs	F (Clinics)	(%)
Availability of diagnostic tools	*Reliance on basic tools only (stethoscope, no x-ray/labs/meds)*	*4*	*100 %*
Training & knowledge	*Uneven auscultation/digital literacy skills*	*4*	*100 %*
Environmental & infrastructural limitations	*Poor signal, noise, overcrowding, lack of supplies*	*4*	*100 %*
Time constraints	*Patient load managed through teamwork*	*3*	*75%*
Referral systems & access	*Overcrowded hospitals, delays, strict requirements, financial barriers*	*4*	*100 %*

All four clinics (100%) reported reliance on basic diagnostic tools, primarily stethoscopes, with no access to x-rays, laboratory tests, or essential medications. This diagnostic gap limits early detection and accurate treatment. Training deficiencies were also universal (100%), particularly in auscultation and digital literacy, suggesting the need for targeted capacity-building. Environmental constraints, poor signal, noise, overcrowding, and supply shortages, were consistently present (100%), compounding operational challenges. Time constraints affected 75% of clinics, with teamwork used to manage patient load, though this strategy may be unsustainable under increasing demand. Referral systems were uniformly strained (100%), with barriers including hospital overcrowding, procedural delays, financial limitations, and strict requirements. These findings highlight systemic limitations in rural healthcare delivery and underscore the need for scalable interventions such as portable diagnostic technologies, structured training programs, and streamlined referral pathways to improve access and equity.

Table 3. *Summary of Prioritized Design Features among Barangay and Satellite Health Workers*

Category	User need/ Preference	Essential Insights
Form Factor	*Ergonomic portability;*	*Health workers prefer a compact, lightweight, and*

	Pediatric usability	*pocket-sized device with retractable or longer tubes suitable for children.*
Interface Layout	*Visual simplicity; Decision support*	*The GUI should have clear buttons, icons, and visual cues to aid quick judgment and reduce misinterpretation.*
Portability	*Field readiness; Reliable power*	*The device must be portable, durable, and usable during emergencies, with rechargeable or flexible power sources.*
Maintenance & Cleaning	*Hygiene-ready design; Reprocessing*	*Health workers emphasized waterproof, rust-resistant materials that can be easily disinfected or safely disposed of.*
Workflow Compatibility	*Seamless embed; Offline reliability*	*Integration with telemedicine and offline functionality ensures easy adoption in remote barangays.*
Cost & Affordability	*Cost sensitivity; Price barrier*	*Devices priced below ₱2,000 are preferred due to tight barangay budgets, ensuring accessibility and sustainability.*

Table 3 revealed that barangay and satellite health workers prioritize a device that is compact, user-friendly, and affordable, reflecting the realities of low-resource, community-based healthcare. Consistent with Ginsburg et al. (2021), portability and ergonomic design are essential to enable use in remote or emergency contexts. Workers value simple interfaces and visual clarity, aligning with Rowe et al. (2020), who emphasized usability for varying literacy levels. Ease of cleaning and durability are critical for infection control (World Health Organization, 2021), while offline compatibility supports telemedicine integration amid unstable internet connectivity (Harrison & Karim, 2019). Finally, cost sensitivity remains decisive, as affordability directly influences adoption in barangay settings (Adebayo et al., 2020). Collectively, these findings highlight the need for an accessible, robust, and context-tailored screening device that complements existing workflows and empowers frontline workers to improve respiratory disease detection at the primary care level.

Figure 4. *Design Feature Prioritized by Health Workers*

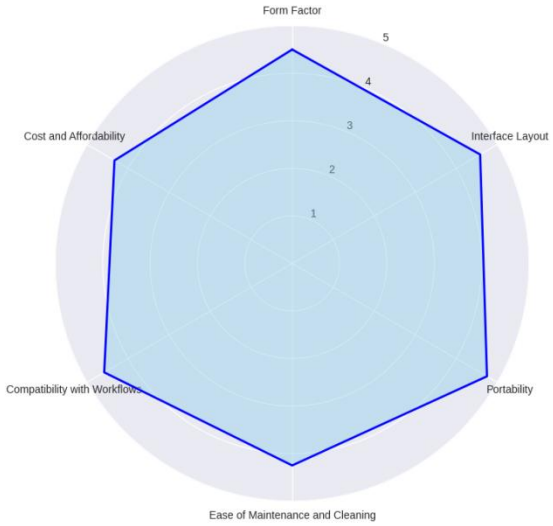


Figure 4 highlights portability (avg ≈ 4.9) as the most critical design priority, underscoring the need for long battery life, ruggedness, and ease of transport in resource-limited clinics. This aligns with Weichelt et al. (2024), who found that reliability and portability are decisive for mHealth adoption in rural care. Interface layout (avg ≈ 4.8) was also emphasized, with simplicity and clear iconography improving usability, consistent with ergonomic design principles outlined by Seo et al. (2024), who stressed clarity and intuitive interaction as determinants of device acceptance. Workflow compatibility (avg ≈ 4.7) reflects health workers' demand for seamless integration, echoing findings by Gabiola et al. (2024) that mobile clinics succeed when tools fit existing patient flows. While affordability (avg ≈ 4.4) remains important, respondents prioritized cost-benefit balance over low price alone, resonating with Intertek (2025) on accessibility through user-centered design. Maintenance and cleaning (avg ≈ 4.3) were valued but less dominant, suggesting usability and reliability outweigh technical upkeep in frontline contexts.

Table 4. *Frequency of User Preferences for Screening Device Design*

Category	**User Needs**	**F (Clinics)**	**(%)**
Form factor	*Compact, protective, ergonomic*	*3*	*75%*
Interface layout	*Simple, visual, icon-based, with training support*	*4*	*100 %*
Portability	*Battery life, ruggedness, ease of transport*	*2*	*50%*
Ease of maintenance & cleaning	*Waterproof, disinfectable, protective layers*	*4*	*100 %*
Workflow compatibility	*Integration without burden, telemedicine, ready-to-use*	*4*	*100 %*
Cost & affordability	*Low-cost, barangay budget-friendly*	*3*	*75%*

Table 4 highlights the user preferences across clinics reveal consistent design priorities for diagnostic tools. All clinics (100%)

emphasized the need for a simple, visual, icon-based interface with built-in training support, highlighting the importance of intuitive usability in low-resource settings. Ease of maintenance and cleaning was also universally preferred (100%), with waterproofing, disinfectability, and protective layers seen as essential for hygiene and durability. Workflow compatibility—including telemedicine integration and ready-to-use functionality—was prioritized by all clinics (100%), underscoring the need for seamless adoption without added burden. Compact, ergonomic form factors and affordability were preferred by 75% of clinics, aligning with barangay-level budget constraints and space limitations. Portability features such as battery life and ruggedness were noted in only 50% of clinics, suggesting variability in mobility needs. These findings inform the design of context-sensitive health technologies, emphasizing simplicity, hygiene, affordability, and integration over advanced portability, and guiding future development toward scalable, community-ready solutions.

Figure 5. *Performance Metrics for Evaluating SmartSteth's Efficiency*

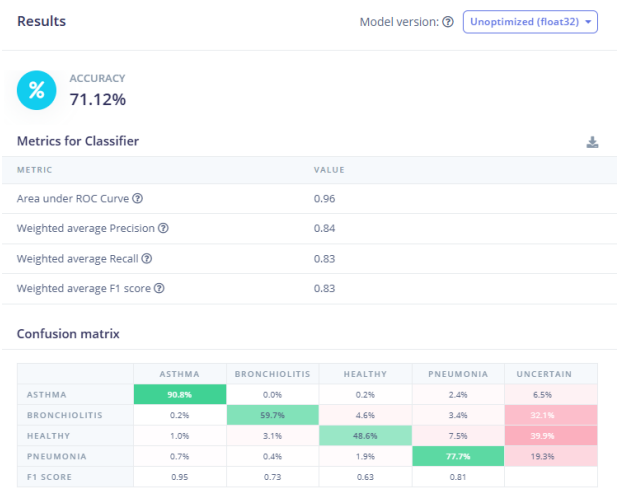


Table 5. *Key Performance Metrics for Evaluating SmartSteth's Efficiency*

Metric	Evaluation Method	KPI / Target	Relevance
Signal-to-Noise Ratio (SNR)	*Bench and field recordings under variable noise*	*≥20 dB (quiet), ≥10 dB (field)*	*Ensures clear auscultation and accurate AI detection*
Frequency Resolution	*FFT analysis (50–2000 Hz)*	*≥5–10 Hz resolution*	*Enables precise identification of respiratory sounds*
Latency	*Time from input to AI output*	*2 ms*	*Supports real-time decision-making*
Sensitivity to Sound Types	*Comparison with expert-labeled dataset*	*≥70% sensitivity; F1 ≥ 0.83*	*Indicates diagnostic reliability*
Robustness to Noise	*Controlled and field noise testing*	*≥80% sensitivity at 5–10 dB SNR*	*Confirms usability in real environments*
Consistenc Across Profiles	*Stratified performance analysis*	*≤10% variance*	*Promotes equity and generalizability*

Table 5 and Figure 5 reveal that SmartSteth's efficiency in respiratory sound

screening relies on achieving an optimal balance between signal clarity, diagnostic sensitivity, and robustness under real-world conditions. A high signal-to-noise ratio (≥20 dB) and frequency resolution of 5–10 Hz enable accurate detection of respiratory sounds such as wheezes and crackles, aligning with recent advances in digital auscultation and deep-learning stethoscope systems (Huang et al., 2023; Lee et al., 2025). Low latency (2 ms) supports real-time clinical use, consistent with the performance of wearable stethoscopes designed for automated diagnosis (Science Advances, 2024). Robustness at 5–10 dB SNR addresses ambient noise challenges frequently observed in community and low-resource settings (Journal of Global Health, 2025; Lee et al., 2025). Maintaining ≥70% sensitivity and an F1-score ≥ 0.83 across diverse sound types ensures diagnostic reliability comparable to current state-of-the-art AI classifiers (Xie, Zhou & Xu, 2025). Moreover, consistent performance across pediatric and adult profiles promotes inclusivity and diagnostic fairness, essential for equitable healthcare delivery.

Table 6. *Summary of Frontline Health Workers' Perceptions on GUI Intuitiveness*

Category	Essential Theme	Essential Insights
Ease of Navigation	*Straightforward interaction*	*Buttons and menus should be clearly labeled and logically arranged for quick use during patient consultations.*
Clarity of Instructions	*Embedded guidance*	*Built-in user or video guides help minimize confusion, support independent learning, and improve operational accuracy.*
Accessibility (Low Literacy)	*Inclusive interface*	*The GUI must accommodate varying digital literacy levels through simple layouts, icons, and optional customization.*
Error Prevention	*Conservative outputs*	*The app should emphasize that results are predictive or supportive, not diagnostic, to maintain clinical safety and trust.*
Training Time	*On-demand learning*	*Short, accessible modules such as webinars and app-based tutorials promote proficiency among busy or remote health workers.*

		Providing multiple language options can enhance accessibility, especially for barangay-based users across diverse regions.
Multilingual Support	*Language inclusivity*	

Table 6 reveals that frontline health workers value a simple, clear, and supportive GUI that reduces cognitive load and accommodates varying digital literacy levels. Participants emphasized clear navigation and labeled buttons (MC_P01), embedded video or written guides (MC_P01), and options suitable for senior or less tech-savvy users (SC_P01). These preferences mirror findings in recent usability research, which highlight clarity, consistency, and error prevention as key determinants of user satisfaction in mobile health interfaces (Lim et al., 2025). Similarly, an umbrella review underscored the importance of plain language, visual cues, and intuitive navigation in improving accessibility for low-literacy users (BMC Medical Informatics & Decision Making, 2024). Evidence from wearable health technologies in low-resource contexts also stresses the value of training support and user-friendly layouts for adoption and trust (Nakaggwa et al., 2024). Collectively, these affirm that SmartSteth's GUI must prioritize simplicity, guidance, and inclusivity to ensure effective frontline use.

Figure 6. ***GUI Usability Analytics (Heuristic Evaluation)***

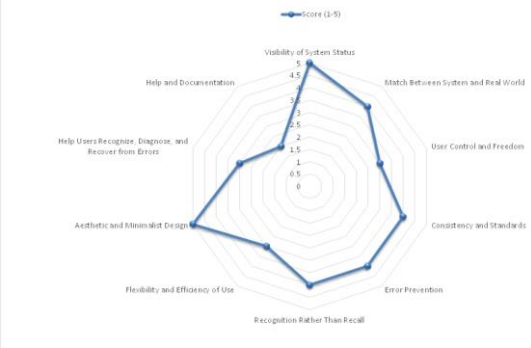


Figure 6 underscores that SmartSteth's acceptance and trust are shaped by six interlinked factors: perceived reliability, cultural fit, transparency, prior digital experience, community endorsement, and support infrastructure. Reliability and consistent diagnostic performance were most emphasized, aligning with Tun et al. (2025), who found that explainability and accuracy are foundational to trust in AI diagnostics. Cultural compatibility and community relevance also emerged as key, echoing Nakaggwa et al. (2024), who demonstrated that culturally anchored mHealth tools foster deeper engagement in rural settings. Transparency, especially understandable outputs and clear AI result explanations, reinforces ethical integration, consistent with O'Sullivan et al. (2024) on clinician-AI collaboration. Community leader endorsement and peer influence further support adoption, while short, practical training and ongoing support were vital for digitally novice users, reflecting Lim et al. (2025), who identified training accessibility as a usability determinant. Together, these findings suggest that SmartSteth's success depends on balancing technical reliability with social, cultural, and educational responsiveness.

Table 7. *Graphical User Interface Features Evaluation of the SmartSteth Web App*

Factor (GUI Feature)	*F*	%
Ease of Navigation (clear buttons)	1	25%
Clarity of Instructions (guides, video tutorials)	1	25%
Accessibility (variable digital literacy)	1	25%
User Options & Flexibility (posting options, simplified tasks)	1	25%
Error Prevention & Risk Communication (non-diagnostic framing)	2	50%
Training & Capacity Building (short, flexible training)	2	50%

N = 4

Table 7 indicates that frontline health workers expressed varied but converging preferences for a graphical user interface (GUI) that is simple, supportive, and adaptable to different levels of digital literacy. Individual participants emphasized distinct needs: clear navigation and labeled buttons (MC_P01), embedded video or written guides (MC_P01), options for less tech-savvy or senior users (KI_04), and customizable posting features (SC_P01). Two participants underscored the importance of non-diagnostic framing to prevent misuse (MC_P01, KI_04) and flexible training formats such as webinars (SV_P01, MC_P01). Although consensus was limited in this small sample, the themes collectively highlight the importance of reducing cognitive load, providing guidance, and accommodating heterogeneous user skills. These findings align with broader evidence that clarity, consistency, and error prevention drive usability in mobile health applications (Lim et al., 2025), while plain language, visual cues, and training support improve accessibility and adoption in low-resource contexts (BMC Medical Informatics & Decision Making, 2024; Nakaggwa et al., 2024).

Table 8 . *Factors Influencing Acceptance and Trust in SmartSteth*

Category	Essential Theme	Essential Insights
Perceived Usefulness & Reliability	*Conditional trust; Adjunct utility*	*Health workers value SmartSteth as a decision-support tool that enhances assessment accuracy, but long-term trust depends on consistent AI performance and validated results.*
Cultural Compatibility & Relevance	*Adaptation to local practice*	*Acceptance relies on gradual transition from manual to digital methods and clear communication of the device's role within diverse cultural and community settings.*

Previous Experience with Digital Tools	*Familiarity-driven confidence*	*Users with prior exposure to digital or telemedicine tools express higher trust and willingness to integrate SmartSteth into their workflow.*
Community Endorsement & Peer Influence	*Social validation*	*Endorsements from local leaders, medical missions, or peer demonstrations strengthen confidence and encourage adoption among barangay workers.*
Transparency of Diagnostic Results	*Role clarity; Clear scope*	*Explicitly stating that SmartSteth provides supportive—not definitive—findings fosters responsible use and maintains respect for physician authority.*
Support & Training Provided	*Structured capacity-building*	*Continuous access to webinars, manuals, and orientation sessions empowers users, promotes confidence, and reinforces sustained acceptance.*

Table 8 reveals that SmartSteth's acceptance and trust hinge on reliability, transparency, cultural fit, and ongoing training. Health workers emphasized the need for consistent AI accuracy and contextual validation before integrating new tools, consistent with findings by Tun et al. (2025), who reported that reliability and explainability are core to trust in AI-driven diagnostics. Transparent communication of results and maintaining physician oversight further support ethical use, echoing the conclusions of O'Sullivan et al. (2024) on clinician-AI collaboration. Cultural alignment and community endorsement remain critical, as supported by Nakaggwa et al. (2024), who found localized outreach essential for mHealth adoption in low-resource settings. Finally, structured, short-format training improves confidence among digitally novice users, aligning with Lim et al. (2025), who highlighted training accessibility as a determinant of usability and trust in mobile health applications.

Figure 5. *Stacked Bar Chart of the Factors of User Acceptance and Trust*

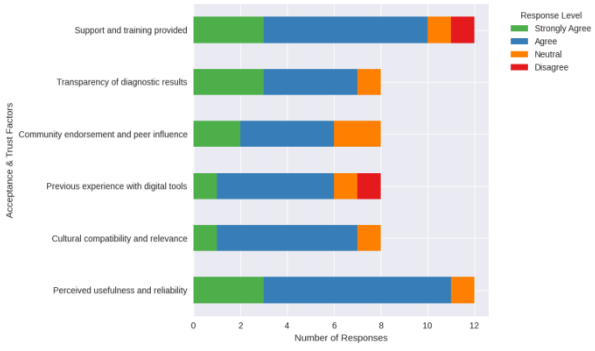


Figure 5 demonstrates that SmartSteth's acceptance and trust are shaped by perceived reliability, transparency, cultural fit, and training support. Reliability emerged as the

strongest determinant, with health workers emphasizing consistent diagnostic performance, echoing Tun et al. (2025), who identified accuracy and explainability as central to trust in AI-driven diagnostics. Transparency of outputs and physician oversight further reinforce ethical integration, consistent with O'Sullivan et al. (2024), who highlighted clinician-AI collaboration as vital for safe adoption. Cultural compatibility and community endorsement also play a decisive role, aligning with Nakaggwa et al. (2024), who found that localized outreach and cultural relevance are essential for Health acceptance in low-resource contexts. Training and ongoing support were valued, particularly short, practical sessions that build confidence among digitally novice users, reflecting Lim et al. (2025), who emphasized training accessibility as a key determinant of usability. Together, these findings suggest that SmartSteth's adoption depends on balancing technical reliability with cultural and social integration.

Table 9. *Summary of Novelty in SmartSteth*

Feature	CN119296591A	WO2025085557A1	SmartSteth
Target Organs	*Heart + Lung*	*Heart + Lung*	*Lung only*
Signal Processing	*SpecAugment + Mel*	*Visual + Audio*	*Mel-spectrogram + basic denoising*
Hardware	*Proprietary device*	*Proprietary device*	*ESP32 + MEMS mic + acoustic scope*
Deployment	*Embedded*	*Embedded*	*Web-based (for now)*
User Context	*Clinical*	*Clinical*	*Rural frontline health workers*
Patent Status	*Filed (CN)*	*Filed (WIPO)*	*Not yet patented*

Table 10. *Comparative assessment of SmartSteth and E-Steth across key design and usability categories*

Categories	SmartSteth	E-Steth
Interface and Usability	*Simple, icon-based interface; easy for low-digital users.*	*Requires internet and higher digital literacy.*
Portability and Offline Use	*Pocket-sized, rechargeable, works offline.*	*Needs stable internet; less field-ready.*
Maintenance and Cleaning	*Waterproof, rust-resistant, easy to disinfect.*	*Harder to clean; complex build.*
Fit with Existing	*Fits barangay workflows;*	*Designed for hospitals;*

Workflows in Rural Clinics	*supports offline use.*	*network-dependent.*
Cost and Affordability for Community Health Facilities	*Costs under ₱2,000; locally built and affordable.*	*Expensive (₱15k–₱40k); less accessible.*

Table 9 and 10 highlights SmartSteth's distinct advantages in both technological novelty and contextual fit for rural healthcare. Unlike CN119296591A and WO2025085557A1, which target both heart and lung sounds using proprietary embedded systems, SmartSteth focuses solely on lung auscultation using a low-cost ESP32 and MEMS microphone setup. Its signal processing is simplified, using Mel-spectrograms and basic denoising, yet effective for respiratory classification. While existing patents are designed for clinical environments, SmartSteth is tailored for frontline health workers, operating via a web-based interface with future offline deployment planned. Usability comparisons with E-Steth further emphasize SmartSteth's strengths: it offers a simple, icon-driven interface, offline functionality, and easy maintenance, all at a fraction of the cost (under ₱2,000). E-Steth, by contrast, requires stable internet, higher digital literacy, and is less suited to barangay workflows. Collectively, these tables demonstrate SmartSteth's potential as a scalable, locally adaptable, and affordable diagnostic tool for underserved communities.

IV. CONCLUSION

SmartSteth proves that artificial intelligence can be adapted to meet the needs of community-based healthcare. By combining CNN-driven lung sound analysis with a compact and intuitive design, the device helps health workers detect respiratory conditions more accurately and confidently. Its offline functionality, low cost, and user-friendly interface make it suitable for remote clinics and mobile health missions. The study also highlights the importance of cultural fit, training support, and transparent communication in building trust and promoting adoption. As health systems move toward digital transformation, SmartSteth offers a practical and scalable solution for improving respiratory care in low-resource settings.

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APPENDIX A

RAW DATA (RESULTS)

Interview transcript – Sicsican Clinic

- **Researchers:** Kiesha Mekhaela B. Relatado, Teokislev Gazelle J. Jompella
- **Participant code:** SC-P01
- **Date of interview:** September 30, 2025
- **Location of interview:** Barangay Sicsican Clinic

List of acronyms:

- **AI** — Artificial Intelligence
- **CHO** — City Health Office
- **URTI** — Upper Respiratory Tract Infection
- **BP** — Blood Pressure
- **IT** — Information Technology

Transcript

(00:05)

Kiesha Mekhaela B. Relatado: Hello po. Yung age, optional lang po. Gagamitin po namin yung pangalan niyo sa research.

Kiesha Mekhaela B. Relatado: What is your age?

SC-P01: 34.

Kiesha Mekhaela B. Relatado: Profession?

SC-P01: Nurse.

Kiesha Mekhaela B. Relatado: Years of experience in healthcare?

SC-P01: Almost 4 years.

(00:43)

Kiesha Mekhaela B. Relatado: Yung study po namin ay tungkol sa SmartSteth. AI-powered stethoscope po siya na nagre-record ng lung sounds at nagsa-submit sa website para ma-predict ang most likely respiratory illness.

Kiesha Mekhaela B. Relatado: Target po namin ay rural areas, kaya po kami nandito sa Barangay Sicsican. User acceptance po ang focus ng interview.

(01:32)

Kiesha Mekhaela B. Relatado: What difficulties do you face when identifying respiratory symptoms in the community?

SC-P01: Depende sa brand ng stethoscope at sa positioning. Pero mostly, okay naman.

(02:00)

Kiesha Mekhaela B. Relatado: How does limited training or knowledge affect your confidence in recognizing these diseases?

SC-P01: Para sa iba, possible na mamisinterpret ang findings—lalo na kung hindi masyadong naririnig ang lung sounds.

(02:37) **Kiesha Mekhaela B. Relatado:** How do time constraints or environmental barriers like poor ventilation or lack of electricity affect diagnosis?

SC-P01: Wala namang malaking difference. Pero kapag maingay ang paligid, doon nagkakaproblema—may ibang sounds na pumapasok sa stethoscope.

(03:13)

Kiesha Mekhaela B. Relatado: What device features would make it easier to use in the field—like simplicity, clarity, icon-based interface?

SC-P01: Since rural ang target, dapat offline-capable. Maraming lugar walang internet.

(03:46)

Kiesha Mekhaela B. Relatado: As of now, online pa siya. Pero plan po namin na maging offline-capable ang device.

SC-P01: Okay. Kasi sa far-flung areas, mahirap talaga pag walang internet. Manual pa rin ang gamit doon.

(04:35)

Kiesha Mekhaela B. Relatado: Kung ima-mass produce po namin ang device, sa tingin niyo po matatanggap siya ng medical community?

SC-P01: Possible. Makakatulong siya sa medical professionals. Pero dapat consistent ang AI updates.

(05:15)

Kiesha Mekhaela B. Relatado: Given na maraming Filipinos ang nagse-self-diagnose, how do you encourage families in Sicsican to seek help?

SC-P01: Ina-advise namin sila na mag-consult sa health center. Kung hindi kaya dito, nire-refer namin sa City Health o preferred hospital.

(06:23) **Kiesha Mekhaela B. Relatado:** How important is portability, battery life, and ease of transport in your daily work?

SC-P01: Important siya, lalo na sa emergency cases. Dapat mabilis madala ang pasyente sa facility.

(06:48)

Kiesha Mekhaela B. Relatado: What level of affordability would make the device accessible for Barangay Sicsican?

SC-P01: Medyo tight ang budget ng barangay. Usually, City Health ang nagbibigay ng equipment sa designated midwife.

(07:20)

Kiesha Mekhaela B. Relatado: Sa tingin niyo po, ano ang price range na tatanggapin ng barangay?

SC-P01: Depende sa budget. May sample na ba kayo ng device—like picture or prototype?

(08:19)

SC-P01: Wala pa po as of now. Kakabili lang namin ng microphone sa Manila. Gagawin pa lang yung product.

Kiesha Mekhaela B. Relatado: May design na po ba kayo?

SC-P01: Opo, may picture na kami ng design.

(08:51)

SC-P01: Para may pagpipilian sa posting, para mas madali sa users.

SC-P01: We're maintaining affordability. We're just integrating AI.

(09:26)

Kiesha Mekhaela B. Relatado: Ganito lang talaga siya kaliit?

SC-P01: Opo. Didikit sa patient. Portable. Depende sa microcontroller.

SC-P01: Parang pocket size. Retractable yung tube—nahihila-hila.

(09:51)

SC-P01: Budget namin sa project is around ₱3,000.

Kiesha Mekhaela B. Relatado: Sa tingin niyo po, ano ang acceptable price range?

SC-P01: Less than ₱2,000. Pero kung marami siyang features, okay lang. Basta consistent ang updates.

(10:39)

Kiesha Mekhaela B. Relatado: How do you think the community will respond to a digital diagnostic tool like SmartSteth?

SC-P01: I can explain its purpose to healthcare professionals.

SC-P01: Depende kung magshi-shift sila from manual to digital—like BP apparatus, may manual at digital.

SC-P01: Kung saan comfortable at efficient, doon sila mag-aadjust.

(11:17) **Kiesha Mekhaela B. Relatado:** Sa tingin niyo po, magiging skeptical sila dahil AI-based?

SC-P01: Hindi naman siguro. Diagnosis pa rin ng doktor ang basehan.

(11:40)

Kiesha Mekhaela B. Relatado: What role does cultural compatibility play in acceptance?

SC-P01: Malaking tulong ito.

Kiesha Mekhaela B. Relatado: How important is endorsement from community leaders?

SC-P01: Pwede mag-conduct ng orientation or medical missions para ma-endorse ang device. Para maging aware ang community.

(12:44)

Kiesha Mekhaela B. Relatado: Ano pong advice niyo sa design? Kailangan po ba mahaba ang tube?

SC-P01: Yes. Especially sa children—agitated sila pag natatouch.

SC-P01: Mahaba ang tube para hindi madamage ang device.

SC-P01: Concise size, complete features. May chip or memory card?

(14:04)

Kiesha Mekhaela B. Relatado: Microcontroller po siya—wireless transmission to website.

SC-P01: Okay. Pero kung offline, mas maganda kung may memory card.

Kiesha Mekhaela B. Relatado: Possible po ‘yan.

(14:22)

Kiesha Mekhaela B. Relatado: What are the most urgent needs for improving respiratory disease diagnosis in your setting?

SC-P01: May nagpapacheck-up naman. Mostly URTI, cough and colds.

SC-P01: Babies are hard to auscultate—di mo sila mapapahinga ng malalim.

(15:21)

SC-P01: Kung may noise cancellation, mas makakatulong. Focused sa internal lung sounds.

(15:43) **Kiesha Mekhaela B. Relatado:** Final advice for SmartSteth developers?

SC-P01: Continue improving the device.

SC-P01: After release, conduct pilot testing. Address priority issues first.

SC-P01: Hindi lahat ng problems mare-resolve agad. Piliin ang pinaka-importante.

(16:46)

SC-P01: Depende pa rin sa IT team kung sabay-sabay i-address ang issues.

(17:04)

Kiesha Mekhaela B. Relatado: Prevalent po ba ang respiratory illnesses sa Sicsican?

SC-P01: Depende sa weather. Ubo pag maulan.

Kiesha Mekhaela B. Relatado: What about communicable diseases like asthma or pneumonia?

SC-P01: Minsan, community-acquired pneumonia—lalo na kung exposed sa secondhand smoke.

(17:56)

Kiesha Mekhaela B. Relatado: Wala na pong ibang nurse dito?

SC-P01: Ako lang po. Yung midwife naka-leave.

Kiesha Mekhaela B. Relatado: Kayo lang po may experience sa diagnosis?

SC-P01: For now, yes.

Interview Transcript – Salvacion Clinic

- **Interviewer:** Teokislev Gazelle J. Jompella
- **Participant code:** SV-P01 (Registered Nurse, 25 years old, ~2 years in healthcare)
- **Date of Interview:** October 5, 2025
- **Location of Interview:** Barangay Salvacion Clinic

List of acronyms:

- **AI** — Artificial Intelligence
- **CHO** — City Health Office
- **URTI** — Upper Respiratory Tract Infection

- **BP** — Blood Pressure
- **IV** — Intravenous
- **GUI** — Graphical User Interface

(00:00)

Teokislev Gazelle J. Jompella: Edad niyo po?

SV-P01: Edad? 25.

Teokislev Gazelle J. Jompella: Propesyon?

SV-P01: Registered nurse.

Teokislev Gazelle J. Jompella: Ilang taon na po kayo sa healthcare?

SV-P01: Dalawang taon.

(01:39)

Teokislev Gazelle J. Jompella: Paano niyo po nababalanse ang dami ng pasyente at time constraints sa konsultasyon?

SV-P01: Isa-isa lang ang pasyente. Dumadami sila, pero dahil marami kami sa duty, nagtutulungan kami.

SV-P01: Hindi naman siya burden kahit busy. Sanay na kami, lalo na kung may doktor. Talagang teamwork.

(02:34)

Teokislev Gazelle J. Jompella: Ano po ang mga challenges kapag nagpaparefer kayo sa ospital o espesyalista?

SV-P01: Una, problema sa signal. Pangalawa, maraming requirements ang private hospitals.

SV-P01: Minsan walang available na kama. Kaya kadalasan, sa Palawan Hospital napupunta ang pasyente.

SV-P01: May pagkakataon na may dalawa akong critical na pasyente—isa mababa ang BP, isa posibleng may heart attack.

SV-P01: Wala kaming gamot para itaas ang BP. Wala ring IV fluids. Kailangan pa naming maghanap sa ibang barangay.

SV-P01: Iba-iba rin ang requirements ng bawat doktor bago tumanggap ng pasyente.

(05:16)

Teokislev Gazelle J. Jompella: Ano pa po ang ibang gaps na nakikita ninyo?

SV-P01: Health education pa rin. Kailangan maintindihan ng pasyente at pamilya ang kondisyon.

Teokislev Gazelle J. Jompella: May problema po ba sa pag-inform sa komunidad?

SV-P01: Dahil satellite clinic lang kami, hindi kami lumalabas. Ibang sector ang humahawak niyan.

SV-P01: Kami ang humahawak ng emergency cases dito sa area.

(06:12)

Teokislev Gazelle J. Jompella: Paano po dapat i-design ang device para madaling linisin at i-maintain sa rural clinics?

SV-P01: Dapat madaling linisin—spray o swab na hindi makakaapekto sa microphone.

SV-P01: Mas maganda kung may protective layer ang stethoscope at ang box, lalo na sa paligid ng mic.

SV-P01: Para maiwasan ang maling diagnosis.

(09:42)

Teokislev Gazelle J. Jompella: Anong klaseng interface layout ang makakatulong sa mga nurse na limitado ang digital training?

SV-P01: Education pa rin ang susi. Maganda kung may webinars o face-to-face training.

Teokislev Gazelle J. Jompella: Paano po ma-integrate ang device sa kasalukuyang patient care routines nang hindi nakakadagdag ng burden?

SV-P01: Dapat ready-to-use. Pero sana may offline mode, kasi hindi lahat ng lugar may signal.

SV-P01: Kahit offline, dapat accurate pa rin.

(11:56)

Teokislev Gazelle J. Jompella: Paano nakakaapekto ang dating karanasan mo sa digital tools sa willingness mong gumamit ng smart devices?

SV-P01: Para sa akin, madali lang. Medyo inclined kami sa ganitong tech.

SV-P01: Pero yung mga senior nurses, nahihirapan—lalo na sa internet, logins, passwords. Nakakaubos ng oras para sa kanila.

(13:43–13:58)

SV-P01: Medyo maingat ako. Gusto ko munang matest ang device sa maraming lugar—hindi lang isa. (13:58–14:19)

SV-P01: Para makita kung consistent ang performance. Paano kung mahina ang signal dito? Doon mo malalaman kung talagang reliable ang device.

(14:19–14:44)

Teokislev Gazelle J. Jompella: Paano niyo po ipapaliwanag ang resulta ng device sa pasyente at pamilya para magkaroon sila ng tiwala?

SV-P01: Kung gumagana nang maayos ang device, ipapaliwanag ko ito kasabay ng signs and symptoms ng pasyente.

SV-P01: Base pa rin ako sa physical assessment. Add-on lang ang device sa initial findings.

(15:20–15:49)

Teokislev Gazelle J. Jompella: Sa pagbabalik-tanaw, ano po sa tingin ninyo ang pinaka-kagyat na pangangailangan para mapabuti ang diagnosis ng respiratory diseases sa setting ninyo?

SV-P01: Health education pa rin. Kulang ang outreach ng gobyerno—lalo na sa mga liblib na lugar.

(16:09–16:47)

SV-P01: Pinaka-epektibo ang house-to-house teaching. Nakakatulong din ang flyers.

SV-P01: May mga tribal areas din dito—mahalagang maipaliwanag nang malinaw sa kanila.

(16:47–17:12)

Teokislev Gazelle J. Jompella: Anong payo ang maibibigay ninyo sa mga developer ng SmartSteth para mas maging tanggap at kapaki-pakinabang ito sa komunidad ninyo?

SV-P01: Dapat accessible kahit offline. Dapat accurate ang results. I-promote pa para malaman ng mga tao.

(17:38–18:02)

SV-P01: May mga ganitong device na—tulad ng Lipman. Hindi na bago sa amin. Pero kailangan pa ring palawakin ang kaalaman.

(18:02–18:34)

SV-P01: Kailangan pa ng mas maraming research. Dahil bago pa ito, may duda pa rin. Kailangan ninyong patunayan ang sarili ninyo. Offline access ang susi.

(18:34–18:59)

SV-P01: Hindi dapat nakakaubos ng oras. Dapat may protective layers ang device—lalo na sa toxic cases kung saan natatapon ang gamit.

(18:59–19:22)

SV-P01: Dapat may charger at port. Kung may website, dapat may guiding questions din.

Interview Transcript – Macarascas Clinic

- **Interviewer:** Teokislev Gazelle J. Jompella
- **Participant code:** MC-P01 (Clinic staff)
- **Date of Interview:** September 30, 2025
- **Location of Interview:** Barangay Macarascas Clinic

List of acronyms:

- **AI** — Artificial Intelligence

- **CHO** — City Health Office
- **URTI** — Upper Respiratory Tract Infection
- **BP** — Blood Pressure
- **IEC** — Information, Education, and Communication
- **TB** — Tuberculosis
- **GUI** — Graphical User Interface
- **ONP** — Ospital ng Palawan
- **COPD** — Chronic Obstructive Pulmonary Disease
- **ECG** — Electrocardiogram

(00:17)

Teokislev Gazelle J. Jompella: Ano po yung pinaka urgent or pressing needs po dito na na-encounter nyo sa pagda-diagnose ng respiratory diseases?

MC-P01: Kami po, hindi nagda-diagnose—doktor po ang gumagawa nun.

MC-P01: For check-ups, meron kaming stethoscope, pero wala kaming x-ray.

MC-P01: Basic lang po talaga dito sa clinic. Kung hindi kaya, referral agad sa hospital.

(01:55)

MC-P01: Teamwork po kami. Usually 2–3 ang naka-duty. Hati-hati sa tasks.

MC-P01: May triage kami—pagdating ng pasyente, kinukuha agad ang vitals, tapos forward sa amin.

MC-P01: Kami ang nagsusulat ng complaints, tapos diretso na sa doktor.

MC-P01: Kailangan mabilis, kasi kung mabagal, aabutin ka ng gabi.

MC-P01: May specific task assignments para efficient ang flow.

(03:20)

Teokislev Gazelle J. Jompella: Ano po yung mga challenges nyo pag nagpaparefer sa hospital?

MC-P01: Una, capacity. Apat lang ang hospital—tatlo private, isa public.

MC-P01: Sa private, problema agad ang financial aspect.

MC-P01: Karamihan dito sa poor sector, kaya sa public hospital kami umaasa.

MC-P01: Pero napupuno rin ang provincial hospital, lalo na sa peak season.

MC-P01: Kapag puno ang private, tinatapon nila sa ONP. ONP, being government, tatanggapin kahit puno—pero hindi agad maaasikaso. (04:37)

MC-P01: Hangga't kaya sa satellite level, minamanage namin.

MC-P01: Pero kung critical, wala nang choice—takbo na sa ONP.

MC-P01: ONP can't refuse, pero hindi rin agad maasikaso dahil sa dami ng referrals from other municipalities.

(05:49)

Teokislev Gazelle J. Jompella: Paano nyo po na-address ang community misconceptions?

MC-P01: Sa barangay health centers, sila ang preventive side—kasama ang IEC.

MC-P01: Dito sa amin, kapag admitted na, part na ng management ang education.

MC-P01: Pagdating nila dito, may sakit na sila—so hindi na preventive, kundi curative na.

(06:49)

Teokislev Gazelle J. Jompella: Nagde-develop po kami ng screening or pre-diagnostic device. Paano po dapat i-design ang device para madaling linisin at i-maintain sa rural clinics?

MC-P01: Dapat waterproofed.

(07:11)

MC-P01: Hindi mo alam kung anong klaseng pasyente ang darating—lalo na sa respiratory cases. May TB patients na sumusuka ng dugo.

MC-P01: Dapat waterproof ang material. May plema, dugo—hindi mo alam.

MC-P01: Mahirap kung absorbent ang material. Hindi puwedeng i-autoclave ang microphone. Hindi rin puwedeng hugasan.

MC-P01: Dapat rust-resistant. Basic alcohol lang ang disinfectant.

MC-P01: Mas okay kung may plastic cover—parang sa ultrasound probe. Pero dapat walang effect sa sound quality.

MC-P01: Infection control ang priority sa medical devices.

(09:19)

Teokislev Gazelle J. Jompella: How important is the interface for nurses with limited digital training?

MC-P01: Pwede ba siya sa cellphone?

Teokislev Gazelle J. Jompella: May microcontroller po sa loob, tapos wirelessly connected sa website.

MC-P01: Maganda kung app-based. Mas accessible.

MC-P01: Pero dapat may video guide—lalo na sa staff na hindi tech-savvy.

MC-P01: Hindi lahat makaka-attend ng training. Sa amin, walo kami pero hindi sabay-sabay.

MC-P01: Mas okay kung may user guide sa app.

MC-P01: Younger staff prefer reading, pero older staff hindi na nagbabasa.

(11:53)

Teokislev Gazelle J. Jompella: How can the device be integrated into current patient care routines?

MC-P01: Hindi siya burden. Part ng basic assessment ang auscultation—lalo na kung may ubo o sipon.

MC-P01: Mas maganda kung may mic na nagre-record ng sound.

MC-P01: Helpful kung puwedeng i-download at i-send sa doktor via Messenger—lalo na kung walang doktor on-site.

MC-P01: Pwede sa telemedicine. May mga digital stethoscope na ganito, pero mahal.

MC-P01: Sana may option to download or share via link—parang iCloud. Kahit link lang, para ma-share ang sound file.

(14:02)

MC-P01: Sound ang importante—crackles, wheeze.

MC-P01: Doktor ang mas particular sa tunog.

MC-P01: Machine may error risk.

MC-P01: Kahit walang interpretation, basta ma-record ang tunog, kaya na naming i-interpret.

(15:36)

Teokislev Gazelle J. Jompella: Ito po ang main design at testing ng model.

MC-P01: May AI na nag-analyze gamit ang dataset from credible sources.

MC-P01: Pero mabigat ang salitang "diagnosis."

MC-P01: Dapat "early warning" lang—para ma-refer agad sa doktor.

MC-P01: Halos magkakapareho ang lung sounds ng pneumonia, asthma, etc.—kailangan pa rin ng x-ray.

(16:34)

Teokislev Gazelle J. Jompella: Model shows confidence level based on 20,000 audio recordings (5 seconds each).

(17:12)

Teokislev Gazelle J. Jompella: Output is automatic—e.g., asthma.

MC-P01: Machine compares new audio to dataset and outputs most similar result.

MC-P01: AI analyzes and suggests possible diseases related to wheezing.

MC-P01: Pero hindi dapat umasa agad sa diagnosis.

MC-P01: Minsan may wheeze and crackles combined—iba't ibang interpretation.

(17:21)

MC-P01: AI ang nag-a-analyze. Pero dapat naka-encapsulate lang yung output—suggestive of possible diseases related to wheezing.

MC-P01: Hindi dapat agad i-diagnose as asthma. Minsan may wheeze at crackles combination.

MC-P01: May upper and lower lung sounds—iba-iba. Minsan may "no lung sound" kung puno ng tubig or hangin.

(18:53)

MC-P01: Tama rin na "respiratory audio classifier" ang tawag—hindi "disease classifier."

MC-P01: Pwede lumabas na "wheeze sound detected, may be suggestive of asthma."

MC-P01: COPD, asthma, etc.—magkakapareho ang tunog. Kaya dapat suggestive lang. (19:35)

MC-P01: Maganda ang idea, pero mabigat kung "diagnosis" ang label. Parang ECG—hindi nagsasabi ng sakit, kundi rhythm lang. Kapag abnormal ang rhythm, saka ka pa-e-evaluate.

(20:25)

MC-P01: Suggestion: panel outputs like "wheeze detected," "no sound," etc.

MC-P01: Kahit isa lang ang ma-detect, okay na.

MC-P01: Output like "wheeze sound detected, may suggest asthma, please consult…" is helpful.

(21:16)

MC-P01: Sound detection is key—whizz, crackles, etc.

MC-P01: Whizz = asthma. Crackles = parang tunog ng hair scratch.

MC-P01: GUI is helpful—buttons should be clear. It's another way to confirm findings.

(22:49)

MC-P01: Pwede bang iwan lang ang stethoscope sa pasyente? Traditional method requires presence.

MC-P01: Pwede ipatong lang, pero kailangan pa rin i-guide ang pasyente.

MC-P01: Machine can compile 5-second audio chunks and compute results.

(23:49)

MC-P01: Mas okay kung less contact—lalo na sa infectious cases. Parang sa COVID, may patch-style stethoscope.

MC-P01: Pwede rin removable or detachable parts—para lusot sa contamination risk. (25:20)

MC-P01: Material ng vest dapat waterproof—like nylon. Pwede i-connect sa cords, detachable setup.

Interview Transcript – San Rafael Clinic

- **Interviewer:** Kiesha Mekhaela B. Relatado
- **Participant code:** KI_04 (Registered midwife; Registered nurse; 30 years old; ~8 years in healthcare)
- **Date of Interview:** September 30, 2025
- **Location of Interview:** Barangay San Rafael Clinic

List of acronyms:

- **AI** — Artificial Intelligence
- **CHO** — City Health Office
- **URTI** — Upper Respiratory Tract Infection
- **BP** — Blood Pressure
- **GUI** — Graphical User Interface
- **IEC** — Information, Education, and Communication
- **IP** — Indigenous Peoples

Transcript

(0:00–0:08)

Kiesha Mekhaela B. Relatado: Sige, magsimula tayo. Puwede ko bang gamitin ang pangalan mo? Optional lang. Kung ayaw mo, okay lang.

(0:09–0:21)

Kiesha Mekhaela B. Relatado: Puwede ba? Okay lang. Edad mo, ma'am?

KI_04: 30.

Kiesha Mekhaela B. Relatado: Propesyon?

KI_04: Registered midwife, registered nurse.

(0:22–0:40)

Kiesha Mekhaela B. Relatado: Ilang taon ka na sa healthcare?

KI_04: Ngayon, 8 years na akong nasa serbisyo. Pero bilang midwife, mahigit 10 years na.

Kiesha Mekhaela B. Relatado: Okay.

(0:41–1:24)

Kiesha Mekhaela B. Relatado: Unang tanong: Ano po ang pinaka-kagyat na pangangailangan na nae-encounter ninyo sa pag-diagnose ng respiratory diseases sa barangay clinic ninyo?

KI_04: Kagyat na pangangailangan? Sa tingin ko, kakulangan sa kagamitan. Dahil nasa rural area kami, hindi lahat may access sa facilities. Kaya ang gamit namin ay basic lang para ma-assess ang pasyente.

(1:25–2:06)

KI_04: Iyon lang ang magagamit namin.

Kiesha Mekhaela B. Relatado: Okay. Paano niyo nababalanse ang dami ng pasyente at time constraints sa konsultasyon?

KI_04: Actually, nababalanse naman... Kapag maraming pasyente, lalo na kung may doktor, naka-duty kami ng 3 araw. Give and take—kung sino ang available, kung sino ang unang hahawak sa pasyente—doon kami nagma-manage.

(2:07–2:20)

KI_04: Timing talaga ang usapan diyan.

Kiesha Mekhaela B. Relatado: Okay. Ano po ang mga challenges sa pag-coordinate ng referral sa ospital o espesyalista?

KI_04: Challenges? Marami.

(2:21–3:03)

KI_04: Sa government facilities, ang challenge ay walang specialist. Kaya kung may pasyente na kailangan ng specific na doktor, minsan hindi nila kaya. Hindi namin sila mapilit—choice nila 'yon, pero malaking issue ang financial capability.

(3:04–3:33)

KI_04: Sa private hospitals, dati kapag tumawag kami para sa referral, okay agad. Pero ngayon, kailangan munang hintayin ang acceptance nila—kahit nasa pagitan ng buhay at kamatayan ang pasyente. Hindi namin sila mapilit kung ayaw tanggapin ng ospital.

(3:34–4:06)

KI_04: Sa mga IPs (Indigenous Peoples), madalas takot sila sa tao o kulang sa kaalaman. Kahit payuhan namin na magpa-transport, mas gusto nilang umuwi.

(4:07–5:03)

Kiesha Mekhaela B. Relatado: Paano niyo hinaharap ang mga maling paniniwala ng komunidad tungkol sa respiratory illnesses?

KI_04: Sa pamamagitan ng health teaching. Sa komunidad, maraming misconceptions. Kaya sa konsultasyon, bukod sa medical advice, nagbibigay kami ng health teaching—kung ano ang dapat gawin, ano ang dapat unahin.

(5:03–5:41)

KI_04: Halimbawa, health teaching kung talagang kailangan ng pasyente.

(5:51–6:19)

Kiesha Mekhaela B. Relatado: Ipapakita namin sa inyo ang TD model ng device namin. Heto siya. Ito ang set—may LED na umiilaw, halimbawa sa upper left. Sa loob, may microcontroller na nakakonekta sa phone app.

(6:19–6:40)

Kiesha Mekhaela B. Relatado: I-develop namin ito base sa advice ng healthcare workers. Gusto nila marinig ang lung sounds sa phone. Ito ang tanong namin.

(6:42–7:28)

Kiesha Mekhaela B. Relatado: Paano dapat i-design ang device para madaling linisin at i-maintain sa clinic ninyo?

KI_04: Katulad ng ibang device—pagkatapos gamitin, i-disinfect. Kung disposable, itapon. Kung reusable, linisin nang maayos.

Kiesha Mekhaela B. Relatado: Okay. Anong layout ng interface ang makakatulong sa mga nurse na limitado ang digital training?

KI_04: Ano ibig sabihin nun?

Kiesha Mekhaela B. Relatado: Ibig sabihin, design ng interface ng device.

(7:28–8:05)

KI_04: Katulad ng mga nurse na nakausap namin—maraming senior nurses ang hindi techy. Hindi ko masabi, kasi sa clinic namin, lahat marunong gumamit. Uma-attend kami ng trainings at seminars. Siguro sa seniors, hindi ko masasabi para sa kanila.

Kiesha Mekhaela B. Relatado: Okay.

(8:23–9:02)

Kiesha Mekhaela B. Relatado: Paano ma-integrate ang device sa kasalukuyang patient care routines nang hindi nakakadagdag ng burden?

KI_04: Puwede naming ipaliwanag ang proseso at advantages ng device. Marunong naman sila—kailangan lang ipaliwanag nang maayos.

(9:02–9:11)

Kiesha Mekhaela B. Relatado: Okay. GUI. GUI.

KI_04: GUI. Oo.

(9:11–9:27)

Kiesha Mekhaela B. Relatado: Ito ang dalawang klase ng device namin.

(9:27–9:45)

Kiesha Mekhaela B. Relatado: Ito ang original design. Para sa testing kung gumagana talaga ang model. Dito makikita ang confidence ng model sa pag-predict ng type ng disease.

(9:46–9:57)

Kiesha Mekhaela B. Relatado: Base sa tunog. Iyon ang pinapakita namin sa model. Dito makikita ang graphical user interface, ang GUI.

(9:58–10:19)

Kiesha Mekhaela B. Relatado: Kapag binago ang screen, makikita ang user interface. Iyon ang karaniwang nakikita ng user kapag binuksan ang website. Halimbawa, kung gagamitin sa pasyente, puwede nitong i-classify kung healthy, may pneumonia, bronchitis, o asthma.

(10:20–10:29)

Kiesha Mekhaela B. Relatado: Depende sa tunog—base sa output ng stethoscope pagkatapos ng auscultation. Tapos may warning.

(10:30–10:36)

Kiesha Mekhaela B. Relatado: Hindi ito final diagnosis. Prediction lang. (10:37–10:46) **KI_04:** Oo, prediction lang. Warning lang. Check-up lang.

(10:46–10:57)

Kiesha Mekhaela B. Relatado: Ganito siya. Base sa dataset, pareho sa tunog sa loob. Iyon ang pinapakita namin sa model.

(10:57–11:04)

Kiesha Mekhaela B. Relatado: Pareho lang. Ito ang GUI. Makikita ang mga buttons.

(11:04–11:11)

Kiesha Mekhaela B. Relatado: Puwede kang mag-upload ng files. Puwede mong i-classify ang audio. Kapag na-classify, lalabas ang percentage.

(11:12–11:20)

Kiesha Mekhaela B. Relatado: Mas madaling maintindihan ng user kasi graphical—hindi lang numbers o words.

(11:21–11:51)

Kiesha Mekhaela B. Relatado: Visual siya. Ito ang pinapakita namin. Gaano ka-accessible ang GUI

APPENDIX B

STATISTICAL DATA (RESULTS)

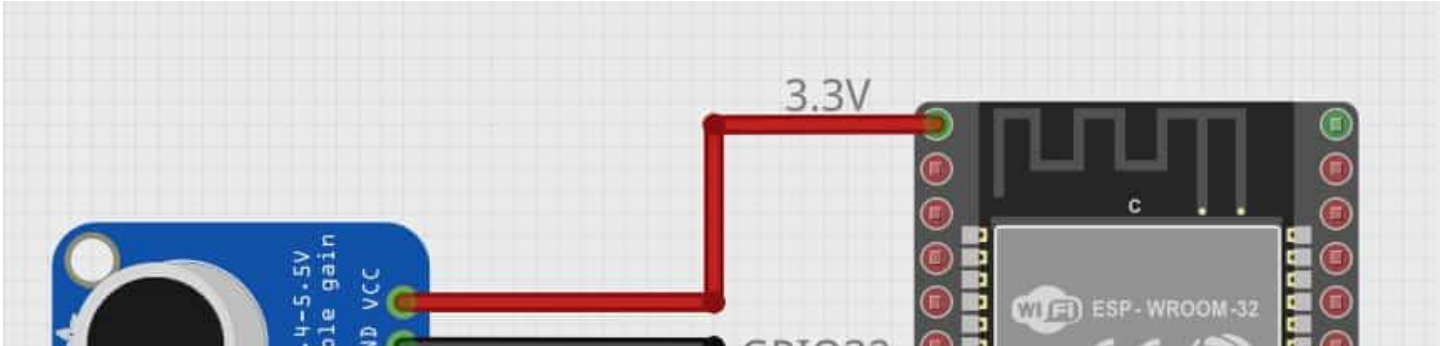
Table 1. Participant Profile Summary

Participant Code	Profession	Gender
SC-PO1	Barangay Nurse	Male
SV-PO1	Satellite Clinic Nurse	Female
MC-PO1	Satellite Clinic Nurse	Female

KI-04	Midwife/Satellite Clinic Nurse	Female

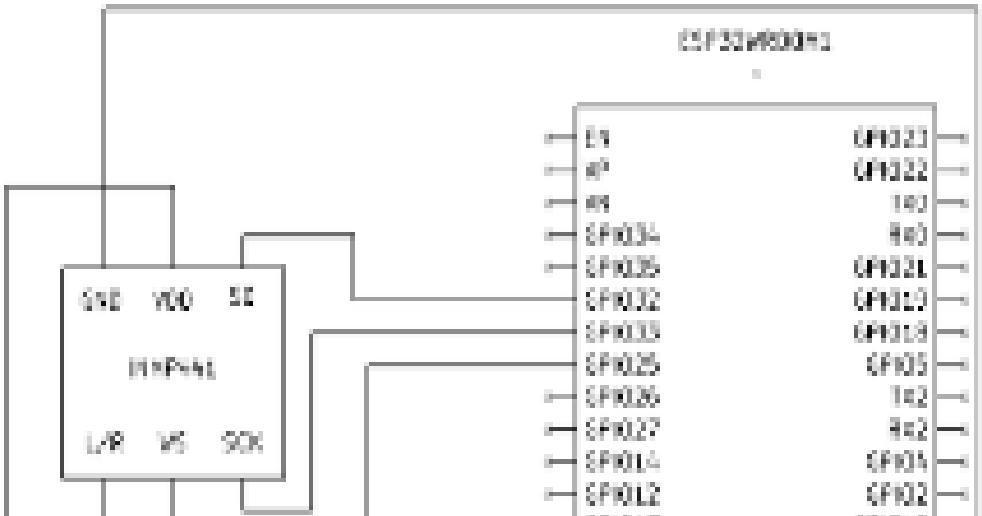
APPENDIX C

SmartSteth Fritzing Diagram



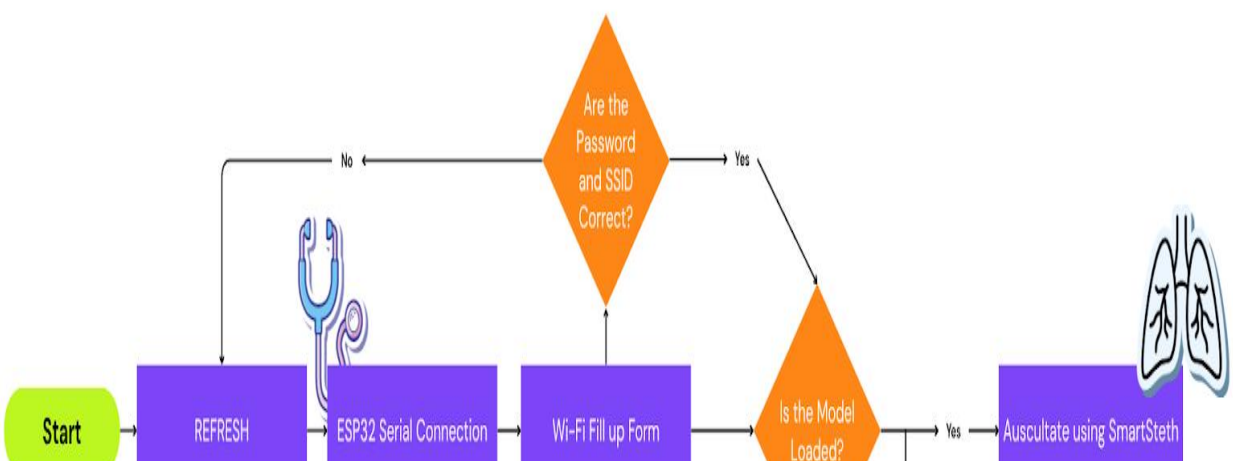
APPENDIX D

Circuit Diagram of the Prototype Device



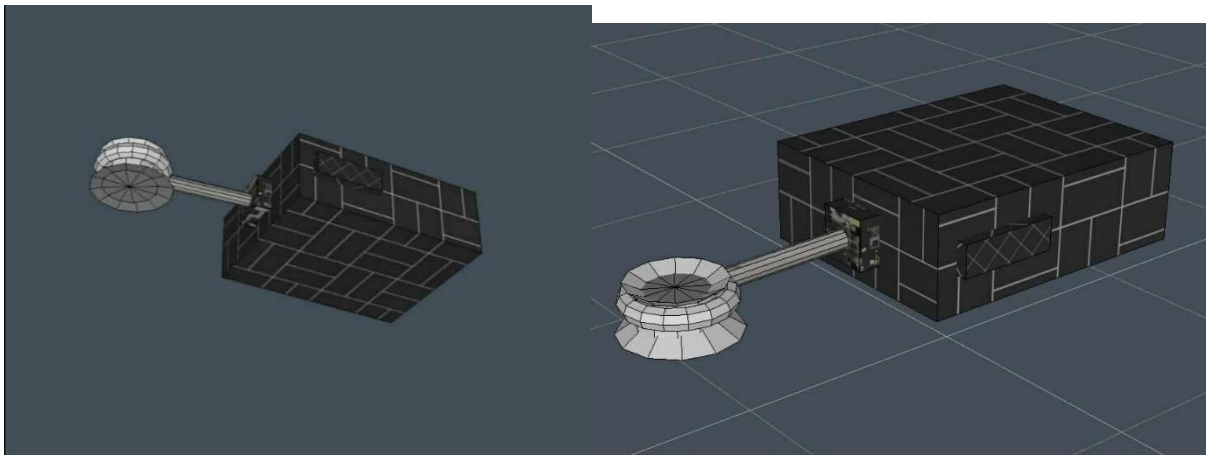
APPENDIX E

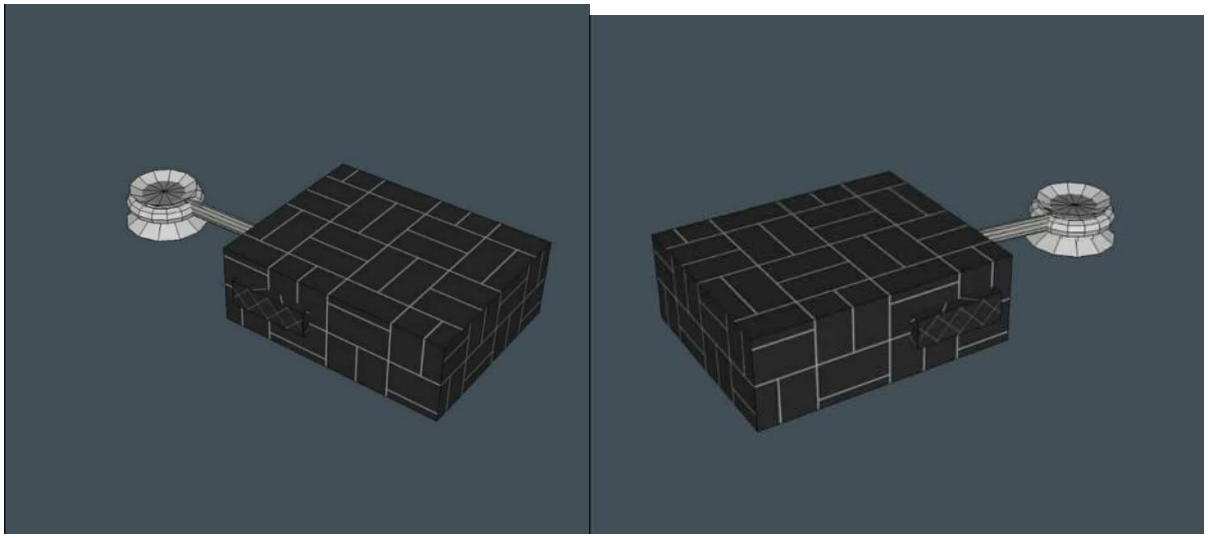
Programing Flow Chart of the Prototype Device



APPENDIX F

3D Model of the Prototype Device





APPENDIX G

Actual Prototype of the device



APPENDIX H

Letters

Republic of the Philippines
Department of Education
MIMAROPA REGION
SCHOOLS DIVISION OF PUERTO PRINCESA CITY

Senior High School Department

October 3, 2025

HON. BALBINO I. PARANGUE
Barangay Chairman

September 30, 2025

Greetings of peace!

We respectfully write to request permission from your office to conduct preliminary interviews with experts from **Barangay Sicsican**. These interviews will support the initial phase of our case-finding and ocular survey for a research study focused on **SmartSteth: A CNN-Driven Auscultation Instrument for Early Detection of Respiratory Diseases (Pneumonia, Tuberculosis, Pertussis) in Rural and Low-Resource Communities Using Respiratory Sound Analysis** undertaken in partial fulfillment of the requirements for Research in Daily Life 1.

This case-finding and interview aims to to record and explore key elements related to the chosen topic, including what techniques are used to detect respiratory communicable disease by voice, how respiratory illnesses are transmitted, how are these illnesses may be detected by voice. By engaging with CHO experts, we hope to gain insights grounded in scientific expertise and community-based knowledge.

This study will utilize interviews as the primary instrument for data collection. The interviews are scheduled to take place tomorrow, with the exact time and venue to be arranged based on the availability and preferences of the participating experts.

In compliance with **Republic Act 10173 – Data Privacy Act of 2012**, we are committed to conducting this research ethically. We will obtain informed consent, ensure participant confidentiality, and prioritize their well-being and autonomy throughout the process.

We sincerely believe that this collaboration will help strengthen research-based education and generate meaningful outputs connected to this sector's development. We respectfully seek your support and favorable consideration for this request.

Thank you very much, and we look forward to your kind response.

Very truly yours,

AHYNNYCA KRISH N. MALIMBAN | **KIESHA B. RELATADO** | **TEOKISLEV GAZELLE J. JOMPELLA**

Researchers/ STEM 11 Rutherford

Noted by:

JOHN LEMUEL J. NOCHE
TIII/ STEM Research Adviser

JANE G. FAVILA
MT II/ Science Cluster Head

Approved by:

DANIEL S. LEBANTE
Principal I

DepED

PUERTO PRINCESA CITY NATIONAL SCIENCE HIGH SCHOOL
Brgy. Santa Monica, Puerto Princesa City, Palawan

N

APPENDIX H



Republic of the Philippines
Department of Education
MIMAROPA REGION
SCHOOLS DIVISION OF PUERTO PRINCESA CITY

Senior High School Department

October 2, 2025

DR. RICARDO B. PANGANIBAN
Medical Officer
City Health Office
Sir:

Greetings of peace!

We respectfully write to request permission from your office to conduct preliminary interviews with experts from the **CITY HEALTH OFFICE (CHO).** These interviews will support the initial phase of our case-finding and ocular survey for a research study focused on **SmartSteth: A CNN-Driven Auscultation Instrument for Early Detection of Respiratory Diseases (Pneumonia, Tuberculosis, Pertussis) in Rural and Low-Resource Communities Using Respiratory Sound Analysis** undertaken in partial fulfillment of the requirements for Research in Daily Life 1.

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Thank you very much, and we look forward to your kind response.

Very truly yours,

AHYNNYCA KRISH N. MALIMBAN
KIESHA B. RELATADO
TEOKISLEV GAZELLE J. JOMPELLA
ARVIN ROGEL A. CORONADO
Researchers/ STEM 11 Rutherford
KHAZZOL R. ZAMORA
ICHIRO N. TOJO

Noted by:

JOHN LEMUEL J. NOCHE
TIII/ STEM Research Adviser

JANE G. FAVILA
MT II/ Science Cluster Head

Approved by:

DANIEL B. LEBANTE
Principal I
October 2, 2025

DepED

PUERTO PRINCESA CITY NATIONAL SCIENCE HIGH SCHOOL
Brgy. Santa Monica, Puerto Princesa City, Palawan

APPENDIX H

Republic of the Philippines
Department of Education
MIMAROPA REGION
SCHOOLS DIVISION OF PUERTO PRINCESA CITY

Senior High School Department

September 26, 2025

MITZI ALMIRA V. GARCIA, BS Information Technology
Facility Manager
Palawan International Technology Business Incubator
Ma'am:

Greetings of peace!

We respectfully write to request permission from your office to conduct preliminary interviews with experts from the **PALAWAN INTERNATIONAL TECHNOLOGY BUSINESS INCUBATOR (PITBI).** These interviews will support the initial phase of our case-finding and ocular survey for a research study focused on **SmartSteth: A Convolutional Neural Network (CNN) Driven Auscultation Instrument Using Lung Sound Analysis for Early Respiratory Disease Detection in Satellite Clinics** undertaken in partial fulfillment of the requirements for Research in Daily Life 1.

This case-finding and interview aims to to record and explore key elements related to the chosen topic, including system architecture, feature extraction methods, model performance variations, data handling protocols, user interface design, and other context-specific applications relevant to the functionality of the voice-based AI device. By engaging with IT experts, we hope to gain insights grounded in scientific expertise and community-based knowledge.

This study will utilize interviews as the primary instrument for data collection. The day, time, and venue of these interviews will be arranged according to the availability and preferences of participating experts.

In compliance with **Republic Act 10173 – Data Privacy Act of 2012**, we are committed to conducting this research ethically. We will obtain informed consent, ensure participant confidentiality, and prioritize their well-being and autonomy throughout the process.

We sincerely believe that this collaboration will help strengthen research-based education and generate meaningful outputs connected to this sector's development. We respectfully seek your support and favorable consideration for this request.

Thank you very much, and we look forward to your kind response.

Very truly yours,

AHYNNYCA KRISH N. MALIMBAN — **KIESHA B. RELATADO** — **TEOKISLEV GAZELLE J. JOMPELLA**

Researchers/ STEM 11 Rutherford

Noted by:

JOHN LEMUEL J. NOCHE
TIII/ STEM Research Adviser

JANE G. FAVILA
MT II/ Science Cluster Head

Approved by:

DANIEL S. LEBANTE
Principal I

DepED

PUERTO PRINCESA CITY NATIONAL SCIENCE HIGH SCHOOL
Brgy. Santa Monica, Puerto Princesa City, Palawan

APPENDIX H



Republic of the Philippines
Department of Education
MIMAROPA REGION
SCHOOLS DIVISION OF PUERTO PRINCESA CITY



Senior High School Department

November 12, 2025

DR. RAMON M. DOCTO
President
Palawan State University
Puerto Princesa City, Palawan

Through: **Intellectual Property and Technology Transfer Office (IPTTO)**

Subject: Request for Assistance in Conducting a Patent Search Report

Dear Sir,

We, the undersigned proponents of the study entitled **"SmartSteth: A Convolutional Neural Network (CNN) Driven Auscultation Instrument Using Lung Sound Analysis for Early Respiratory Disease Detection in Satellite Clinics"**, are humbly requesting your assistance, through the Intellectual Property and Technology Transfer Office (IPTTO), in conducting a **Patent Search Report** for our research.

As Grade 11 students, this study is part of our academic requirement and our commitment to contribute innovative solutions to community health. The Patent Search Report will help us validate the novelty of our proposed invention, ensure compliance with intellectual property standards, and guide us in identifying existing patents or utility models related to our work.

We believe that with the expertise and guidance of the University, particularly through IPTTO, we will be able to strengthen the credibility of our research and uphold the values of responsible innovation.

We sincerely appreciate your time and support in this endeavor, and we look forward to your favorable response.

Respectfully yours,

AHYNNYCA KRISH N. MALIMBAN
Researchers/ STEM 11 Rutherford

KIESHA B. RELATADO

TEOKISLEV GAZELLE J. JOMPELLA

Noted by:

JOHN LEMUEL J. NOCHE
TIII/ STEM Research Adviser

JANE G. FAVILA
MT II/ Science Cluster Head

DANIEL S. LEBANTE
Principal I

Approved by:

Scanned with CamScanner

11/13/25

APPENDIX H

Republic of the Philippines
PALAWAN STATE UNIVERSITY
Office of the Vice President for Research, Development, and Extension
Intellectual Property and Technology Transfer Office
(An IPOPHL-accredited ITSO)
Puerto Princesa City

CERTIFICATION OF PATENT SEARCH

To Whom It May Concern:

This is to certify that the invention titled *"SmartSteth: A Convolutional Neural Network (CNN)-Driven Auscultation Instrument Using Lung Sound Analysis for Early Respiratory Disease Detection in Satellite Clinics"* by **Teokislev Gazelle J. JOMPELLA**, **Ahnycca Krish N. MALIMBAN**, and **Kiesha Mekhaela B. RELATADO**, student-researchers of Puerto Princesa City National Science High School, Puerto Princesa City, is presently undergoing a patent search at the Palawan State University Patent Search Facility.

Issued this 13th day of November 2025 at Palawan State University Main Campus, Tiniguiban, Puerto Princesa City, Philippines.

HERMENEGILDO P. DELA PEÑA
Associate Professor/IPTTO Director

APPENDIX I

Other Documentations



